

Week 1 Readings

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Overview

What is Mesh and LoRa?

- Mesh: a network where devices will directly connect to each other and pass along data for other devices.
- Mesh is decentralized, meaning it generally does not rely on one central router.
- LoRa is a wireless radio technology designed for the long-range communication of low-power devices. Thus, it is commonly used in IoT and mesh communication, where power is a serious constraint.
 - Thus, by nature, LoRa communication is slow.
- LoRa is also resistant to noise (because of Chirp Spread Spectrum) and good at detecting weak signals. However, it is vulnerable to blockage and must have a good line of sight to work effectively.

Why Mesh Communication?

- Programs like Meshtastic and MeshCore use LoRa to create a mesh network, allowing for simple, long-range communication without the use of cellular or Wi-Fi.
- Unlike traditional HAM radio, LoRa does not require certification to use.
- Thus, LoRa mesh communications are highly appealing in the event of an emergency, where all cellular and Wi-Fi communication fails.
 - The most prominent emergency would be a natural disaster. In Halifax, the most likely emergency would be a hurricane.
 - The LoRa frequency band varies based on location. Here, it is 915 MHz.
- Hobbyists enjoy using LoRa mesh communication when hiking, camping or any other 'off-grid' activity, to communicate with friends or family, near or far, without the need to become radio certified.

LoRa Mesh Networking Ecosystems

- The two main LoRa mesh networking ecosystems are Meshtastic and MeshCore.
- Meshtastic is an older, open-source mesh, which uses a flood-based routing (managed flooding) algorithm to communicate. Meshtastic has a much larger community outreach because of its age.
- MeshCore is a newer mesh which uses a role-based, multi-hop routing design to optimize mesh performance when a specific use case is known. MeshCore has a smaller, but rapidly growing community.
- Although Meshtastic vs. MeshCore is a popular debate in the mesh community, the best program depends on the use case.

Meshtastic

- Meshtastic is an open-source communication system that allows low-power devices to communicate long distances without the use of Wi-Fi or a cellular network, by taking advantage of LoRa radio.
- Meshtastic is accessible and easy to use. Anyone can buy a node, flash the firmware onto the device with little restriction and begin communication with other nodes via web or mobile.
- Meshtastic is supported by a wide variety of devices, including most ESP32, nRF52 and Raspberry Pi Devices. A list of supported devices can be found on the Meshtastic documentation.

Meshtastic cont.

- The Meshtastic algorithm is a 'managed flooding' design.
 - See next slide
- Flood routing is a routing method where if a node hears a new packet, it will automatically rebroadcast it until it reaches its destination or until the maximum hop count is reached.
 - Max hops is usually 3
- Thus, every node is both a sender and receiver. For the most part, every node is equal.
 - Nodes that are further away have higher priority to flood first, while closer nodes will refrain from flooding.

Meshtastic cont.

- What makes Meshtastic a managed flooding design is that it uses flood routing, but if another nearby node has already rebroadcasted the current packet, it will not rebroadcast it.
 - Increases simplicity, reduces unnecessary traffic.
- The only exception to the managed flooding design is with direct messaging.
 - If direct messaging is detected, the nodes involved in the previous transmission/response will be marked as next-hop.
- Although Meshtastic nodes do have roles, they mostly change sleep behavior and relay behavior. All nodes still have to participate in flooding.
- Meshtastic packet payloads can be encrypted using AES256-CTR. Header data and metadata will not be encrypted.

Meshtastic LoRa Configuration

- Meshtastic provides a multitude of LoRa configuration settings to allow the user to fine tune the communication based on the mesh setup and use case.
- The most important of these include:
 - Region (Required)
 - Modem Preset
 - Max Hops
 - Transmit Power
 - Bandwidth
 - Spread Factor
 - Coding Rate
- All nodes must share the same configuration settings to communicate.

MeshCore

- MeshCore, just like Meshtastic, is an open-source communication system that uses LoRa to allow low power devices to communicate without the internet.
- Although the two have similar functions, Meshtastic and MeshCore differ in their architecture. While Meshtastic uses managed flooding, MeshCore uses a hybrid next-hop routing system.
- How does hybrid routing work?
 - Flood routing is used for the first message between two nodes. When the message arrives successfully, the receiving node send back a path-return packet. This allows the sending node to add this route to its contacts.
 - For any future communication between the same nodes, the flood will be skipped, and the message will move directly as specified in the saved route.

MeshCore cont.

- Because MeshCore uses a hybrid next-hop routing system, MeshCore requires that nodes have varying roles. MeshCore has three key roles:
 - Companion
 - Repeater
 - Room Server
- Companion devices connect to your phone and are used to send and receive messages. Repeater devices' sole purpose is to rebroadcast packets.
- The downside with the hybrid routing system is that a learned path must stay stable, meaning that if a repeater is moved, it is highly unlikely that the route will break.

Meshtastic vs. MeshCore

- Meshtastic's flooding system can easily create broadcast storms, where there are too many nodes in a mesh and many are rebroadcasting unnecessary data.
- Because of this, Meshtastic has their hop limit set to 7 at most. MeshCore on the other hand, has a hop limit of 64, thus, allowing longer potential routes.
- MeshCore's learned-path routing requires that repeaters do not move. Thus, Meshtastic is the best form of communication for off-grid, mobile groups (For example, a group of hikers or campers)
- Meshtastic also has more supported hardware, documentation and telemetry data.
- Meshtastic is versatile and simple, while MeshCore is best for stationary, optimization-focused meshes.

Readings

Meshtastic Main Site

- The [Meshtastic website](#) is, unsurprisingly, the absolute best resource for anything Meshtastic related.
- The documentation covers Meshtastic basics, as well as the smallest of details.
- Key areas in the documentation include:
 - Radio settings
 - Meshtastic Algorithm and Encryption details
 - Installing and flashing firmware
 - Device configuration (most importantly and radio, module config)
 - Compatible hardware
 - Additional software
 - Development guides
 - And more

MeshCore Main Site

- The [MeshCore main site](#) is an exceptional resource for anything MeshCore.
- However, the documentation of MeshCore's website does not go into nearly as much detail as the Meshtastic site.
- The website docs are one giant FAQ page, covering anything to do with device compatibility, flashing and troubleshooting. It does not go into detail into MeshCore's architecture, algorithm, or encryption.
- In the FAQ, they often delegate answers to questions to third party guides, instead of explaining themselves.

Meshtastic/MeshCore Web Flasher

- The [Meshtastic web flasher](#) is the simplest way to flash firmware onto a device.
- MeshCore has a very similar, easy-to-use [web flasher](#) equivalent to Meshtastic
- The Meshtastic website gives step-by-step instruction how to flash the firmware onto the device, giving different instructions depending on what device is being used.
- The web flasher is a great alternative to the traditional CLI flasher and makes Meshtastic much more accessible.

Meshtastic/MeshCore GitHub

- The [Meshtastic GitHub](#) page includes all of their software, including the firmware, website, Apple and Android apps, web flasher, web client, website API, simulator and more.
- The [MeshCore GitHub](#) includes most their firmware. Notably, MeshCore includes example apps to give a foundation for developers.
- Both are essential resources for building custom firmware, specific documentation and flashing software without the web flasher.

Readings cont.

- [dev.to – First message in 30 minutes](#)
 - The best text tutorial to start using Meshtastic.
 - A simple, easy tutorial with no overcomplication or unnecessary information.
- [Austin Mesh – Getting Started with MeshCore](#)
 - The best text tutorial to start using MeshCore.
 - Like the dev.to tutorial, this tutorial does not beat around the bush, keeping things simple, easy and informative.
- [Kindlymorrow – What is MeshCore](#)
 - A fantastic resource explaining the MeshCore architecture and giving a detailed comparison between Meshtastic and MeshCore.
 - Gives great insight into which ecosystems are thrive depending on the use case

Video Resources

- [**How to Set Up Meshtastic \(Step-by-Step Guide for Beginners\)**](#)
 - Great video for setting up devices and using Meshtastic for the first time.
- [**Web Firmware / Configuration Guide - LILYGO Meshtastic T-Beam V1.1 ESP32 Wireless Module**](#)
 - Best video for showing how to use Meshtastic with the web client.
- [**How to Flash a MeshCore Repeater**](#)
 - Best video for showing how to flash devices with MeshCore. This video shows how to configure a repeater, but setting up a companion is nearly the same.
- [**465 Rutgers University Confirmed: Meshtastic and LoRa are dangerous**](#)
 - More informative video on how to use Meshtastic. Dives into more specifics, including connecting to MQTT servers and integrating sensors nodes into the mesh network. Interesting watch.